TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Core Platform Technology for Inkjet-Printed Stretchable Electrodes and Skin-Conformal Biosensors for Digital Healthcare and Medical Wearables* 디지털 헬스케어 및 의료 웨어러블 기기 응용을 위한 잉크젯 프린팅 기반 신축성 전극 및 피부순응형 바이오신호 센서 핵심 플랫폼 기술			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio and Medicine		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Functional restoration/assistance and welfare equipments & devices	Medical-purpose materials, 500505	High polymer materials	Medical material technology, 400205

Concept of Proposed Technology	• Development of a platform technology for inkjet-printed, stretchable, and skin-conformal electrodes and biosensors.
	  ○ Formulation of conductive–elastic polymer composite inks by combining:
	• Conductive polymers (e.g., PEDOT:PSS) with stretchable polymer binders (e.g., water-borne polyurethane)
	• Targeted simultaneous achievement of:
	  ○ High electrical conductivity			

Rationale for Joint R&D Project	○ Mechanical stretchability ○ Good electrode-skin adhesion ○ Biocompatibility for skin-contact applications • Establishment of industrially relevant inkjet printing processes, including: ○ Stable drop formation and jetting ○ Low temperature drying compatible with polymer and medical substrates • Utilization of non-contact inkjet printing to enable uniform patterning on: ○ Adhesive medical patches and tapes ○ Flexible polymer foils and skin-attachable substrates • Design of electrode and sensor structures with: ○ Stable electrical performance under repeated stretching ○ Robust operation under sweating and temperature variations • Application scope: ○ Wearable heart rate and bio signal monitoring ○ Future extension to medical-grade, skin-conformal biosensing devices
	• Key technical challenges to be addressed: ○ Stable ink formulation for reliable inkjet jetting ○ Control of ink rheology and surface/interfacial properties ○ Low temperature drying and film formation for substrate compatibility ○ Reliable and low-impedance skin–electrode contact ○ Long-term mechanical and electrical stability under body-relevant conditions • These challenges require integrated expertise in: ○ Functional polymer materials science ○ Inkjet printing and process engineering ○ Mechanical, electrical, and environmental reliability assessment • Role of Fraunhofer institute: ○ Advanced design of conductive and stretchable polymer inks ○ Optimization of industrial inkjet printing processes ○ Fundamental and applied reliability evaluation methodologies • Role of Korean SME: ○ Design of stretchable electrode and biosensor architecture ○ Integration of electronics, signal processing, and wireless modules ○ Regulatory compliance, certification, and product commercialization • International joint R&D framework enables: ○ Effective combination of material/process know-how and device/system engineering ○ Systematic technology maturation from TRL 4 (laboratory-scale validation of key materials, components, and processes) to TRL 7 (system-level reliability validation and pre-certification in a relevant operational environment).
Key Words	• Stretchable conductive polymer ink • Inkjet-printed electrodes

	• Skin-conformal biosensors • Wearable bio data monitoring • Flexible medical patches • Heart-rate measurement • Bioelectrical impedance analysis

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	• Establish inkjet-printable conductive and stretchable polymer composite inks with optimized conductivity, stretchability, and biocompatibility. • Develop stretchable electrode and sensor structures that maintain electrical stability under ≥10% cyclic strain and simulated body conditions. • Improve adhesion to skin for good electrical contact without skin sensitization. • Realize low-contact-resistance interconnection concepts between printed electrodes and read-out components (pins, snap buttons, wireless modules). • Verify long-term mechanical and electrical reliability under humidity, temperature cycling, and artificial sweat exposure.			
R&D Target Specifications	• Stretchability: ≥10% cyclic strain for >1,000 cycles with resistance change ≤10% • Electrical performance: Sheet resistance compatible with ECG-grade signal acquisition • Ink properties: Viscosity and surface tension suitable for stable jetting in industrial printheads (e.g., KM1024 class) • Skin–electrode contact: Comparable or improved impedance versus conventional gel electrodes • Environmental stability: Stable performance after exposure to 40 °C / high humidity and artificial sweat			
R&D Plan	1. Material and Ink Formulation a. Optimization of PEDOT:PSS–elastomer blend ratio and doping/dispersion strategy b. Interfacial engineering for balanced conductivity, elasticity, and adhesion 2. Inkjet Printing Process Development a. Control of rheology, drop formation, and satellite suppression b. Low temperature drying and multilayer stacking process windows 3. Stretchable Electrode and Sensor Architecture a. Strain-relief pattern design and electrode–substrate interfacial durability b. Skin-conformal layouts and interconnection structures 4. Reliability and Environmental Testing a. Cyclic stretching, bending, humidity, and sweat simulation b. Long-term electrical and mechanical stability evaluation 5. System Integration Concept a. Interface to signal conditioning, data acquisition, and wireless transmission modules b. Demonstration at wearable prototype level			

	• MS1 (Month 12): Establish stable inkjet-printable stretchable conductive polymer ink and low-temperature printing process with reliable electrical performance under 10% strain. • MS2 (Month 24): Demonstrate skin-conformal stretchable electrode structures with low skin-contact impedance and long-term mechanical and environmental reliability. • MS3 (Month 36): Validate integrated wearable biosensor patch with wireless signal acquisition and pre-certification-level reliability in a real-use environment.

4. R&R (Roles and Responsibilities)

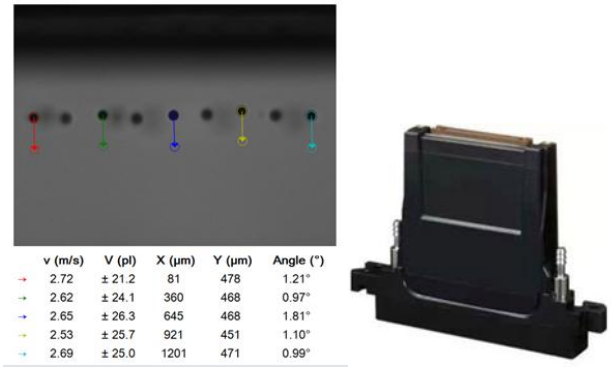
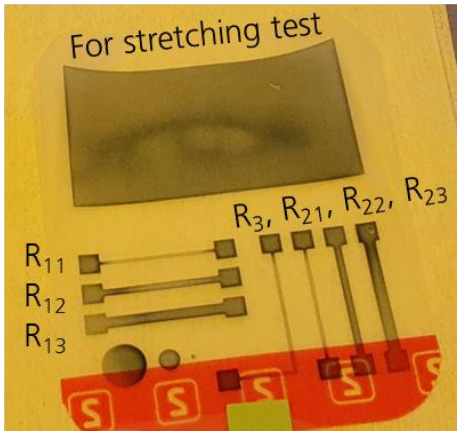
<table>
<tr>
<td>Expected R&R of Research Institute</td>
<td>
Fraunhofer IAP
<ul>
<li>Design of conductive and stretchable polymer composite inks and interfacial property analysis</li>
<li>Optimization of inkjet printing, rheology, drop formation, and drying behavior</li>
</ul>


Fig. Drop-watcher picture demonstrating homogeneous drop formation in the KM1024i-LHE-30pl print head
<ul>
<li>Evaluation of skin–electrode contact resistance, strain–conductivity correlation, and environmental reliability</li>
</ul>


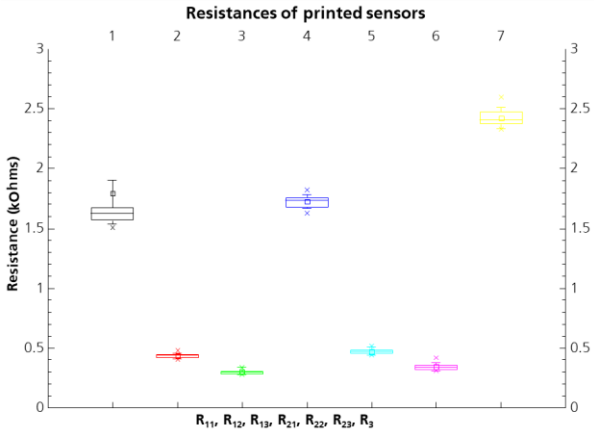


Fig. Example for inkjet-printed test structures on the sticky-side of medical tape and their corresponding resistance. Stretchability is skin compatible (>10 %)
<ul>
<li>Technical support regarding European test standards and pre-CE evaluation methodologies (Medical device certification and clinical approval are not performed by the research institute.)</li>
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<td>Expected R&R of Korean SME Partner</td>
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<ul>
<li>Design of stretchable electrode patterns and biosensor device architecture</li>
<li>Integration of electronics, signal processing, and wireless communication modules</li>
<li>Biocompatibility testing, usability studies, and full regulatory approval (MFDS/CE)</li>
<li>Scale-up, manufacturing process development, quality assurance, and commercialization</li>
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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Technological Outcomes
	  ○ Establishment of an inkjet-printable stretchable electrode and biosensor platform suitable for wearable and future medical-grade applications
	  ○ Securing core technologies in conductive polymer inks, printing processes, and skin-conformal sensor structures
	• Commercialization Potential
	  ○ Development of wearable heart rate and bio signal monitoring patches and next-generation medical sensor modules
	  ○ Strengthening the technological competitiveness of Korean SMEs in printed electronics and biomedical devices
	• Industrial and Economic Impact
	  ○ Creation of high-value-added markets through the convergence of printed electronics and biosensing
	  ○ Promotion of joint IP generation, standardization, and expansion into the European market based on international collaboration

6. References

(1) Bihar, E.; Roberts, T.; Saadaoui, M.; Hervé, T.; De Graaf, J. B.; Malliaras, G. G. Inkjet-Printed PEDOT:PSS Electrodes on Paper for Electrocardiography. Advanced Healthcare Materials 2017, 6(6), 1601167. https://doi.org/10.1002/adhm.201601167

(2) Lo, L.-W.; Zhao, J.; Wan, H.; Wang, Y.; Chakrabartty, S.; Wang, C. An Inkjet-Printed PEDOT:PSS-Based Stretchable Conductor for Wearable Health Monitoring Device Applications. ACS Applied Materials & Interfaces 2021, 13(18), 21693–21702. https://doi.org/10.1021/acsami.1c00537

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization